Conformal Reliability Routing for Low-Storage Few-Shot Industrial Anomaly Detection

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ABSTRACT

Few-shot industrial anomaly detectors built on frozen foundation features can rank defects accurately, but their score thresholds need not retain the same false-alarm meaning across categories or input shifts. We study this reliability problem separately from anomaly ranking. Conformal Reliability Routing (CRR) keeps a low-storage DINOv2 PCA/subspace residual (under 0.6 MB per class) as the ranking score and adds a label-free reliability layer based on leave-one-image-out (LOIO) p-values from k normal support images. We evaluate reliability through fixed-level false-alarm rates, discrete-grid CDF diagnostics, attainable operating levels, and risk–coverage analysis; calibration error is treated as a secondary, prevalence-sensitive metric. On all 12 VisA and 15 MVTec classes under four corruptions, LOIO improves calibration over Platt-family and standard scalar calibrators while preserving ranking. The audit also exposes two limitations: target-only p-values are structurally unable to alarm below $\alpha = 1/(k+1)$, and their empirical false-alarm behavior changes from conservative on VisA to anti-conservative on corrupted MVTec. We therefore introduce Source-Conditioned Cross-Category Reliability Routing (SC3R), which pools normal images from other categories and selects thresholds using held-out source classes. Under a pre-specified evaluation gate, the historical analysis shows near-nominal empirical false-alarm rates at $k = 4$ on MVTec while providing power below the target-only resolution floor. The pattern appears on VisA; one MVTec-to-VisA direction has lower observed false-alarm rates at reduced power, whereas heavier corruptions expose boundary cases at $k = 8$. These results provide empirical evidence for SC3R; population-level target guarantees require explicit source-to-target assumptions and independent certification.

1. Introduction

Industrial anomaly detection (AD) is typically deployed under an asymmetric data regime: normal examples are available, while abnormal examples are rare, diverse, and costly to annotate. This makes few-shot AD practically important. In this setting, an inspection system receives only k normal support images for a target category, often with $k \in \{1, 2, 4, 8\}$, and must detect both image-level and pixel-level defects at test time.

Recent methods have made this regime much more effective by using frozen visual foundation models. DINOv2 [1] patch features can be stored in a normal memory bank and queried by nearest-neighbor distances, as in PatchCore-style methods and AnomalyDINO [2, 3]. Alternatively, frozen patch features can be compressed into a normal PCA/subspace model, as in SubspaceAD [4]. These approaches provide strong anomaly ranking without training a large task-specific backbone.

However, ranking is not the same as reliable decision support. A raw nearest-neighbor distance or PCA residual can produce high AUROC while still being poorly calibrated: a score threshold may work on one class, but the same numerical score may not carry the same false-alarm meaning under another class, dataset, or corruption. This matters in industrial deployment because operators need to know not only which samples are suspicious, but also whether an alarm probability is trustworthy. Prior work has already shown that DINOv2-based few-shot AD can be poorly calibrated and adversarially fragile [5]. Reliability is therefore not a cosmetic post-processing issue; it is part of the deployment problem.

We address this gap by separating anomaly ranking from reliability estimation. The resulting questions are about the learning system itself, not about any one factory: what false-alarm behavior can a decision rule calibrated on k examples of a frozen representation guarantee, at what resolution, and how does that guarantee fail when the representation's inputs shift? The ranking path remains a frozen DINOv2 PCA/subspace detector: patch residuals are aggregated into a raw anomaly score used for AUROC, AP, and localization. A separate reliability path maps this evidence into calibrated or probability-like outputs. We call the resulting framework *Conformal Reliability Routing* (CRR). The main CRR route uses leave-one-image-out (LOIO) conformal calibration over the few normal support images. Each support image is scored as if held out, producing normal nonconformity values; a test image then receives a conformal p-value and a reliability score $1 - p$. This view is not a supervised anomaly posterior, but it is statistically interpretable and derived without target anomaly labels.

Each ingredient we build on—DINOv2 features, PCA residuals, Platt-family calibration, and conformal anomaly detection—has an established literature, and our contribution begins after them. What that literature leaves open is a structural question about the few-shot regime itself: a conformal alarm calibrated on k support images cannot fire below $\alpha = 1/(k+1)$ at all, and corruption shift can make its observed false-alarm rate anti-conservative. The central

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contribution combines an exact resolution-floor diagnosis, an empirical audit under shift, and a source-validated cross-category scheme (SC3R) that alarms below the floor without target anomaly labels. Historical experiments show false-alarm estimates near the pre-specified budget in several evaluated settings; strict nested source certification and target transfer are treated separately. The decoupled detector and calibration benchmark provide a protocol for testing these claims without conflating ranking and reliability.

The lesson for learning systems is general even though the testbed is industrial: the reliability of a few-shot alarm is limited by an exact resolution floor, its empirical failure direction under shift is measurable, and source normal archives can mitigate the resulting operating gap in the evaluated multi-category settings. The paper makes four contributions, the first two central:

- We make the few-shot resolution limit of conformal alarms explicit through an attainable-alpha analysis: with k supports no alarm can fire below $\alpha = 1/(k+1)$, and empirical false-alarm rates must be read against the attainable grid, not the nominal level. Cluster-aware Monte Carlo diagnostics of normal p-values then localize when and in which direction the exchangeable-null pattern fails under corruption shift—conservative on VisA, anti-conservative on corrupted MVTec.
- We introduce SC3R, which pools other-category normals to unlock operating points below the target-only floor. In the historical matched-condition evaluation, a pre-specified empirical gate passes on 15 MVTec classes and replicates on 12 VisA classes. Pointwise class–seed bootstrap intervals support sub-floor power gains, but do not provide simultaneous coverage across conditions. The tested MVTec-to-VisA direction shows reduced power and lower observed false-alarm rates.
- We formulate a decoupled pipeline: frozen DINOv2 PCA residuals provide ranking, while a label-free layer provides Vector/Shift-Aware calibration and LOIO p-values [6] from the normal supports. Across the historical MVTec and VisA corruption studies, LOIO lowers ECE against most tested calibrators without changing the raw ranking; operational metrics, rather than ECE, carry the central interpretation.
- We report accuracy-storage, transfer calibration, pixel-level, robustness, prevalence-stress, and routing diagnostics, including negative findings that prevent overclaiming: CRR is not MVTec AUROC SOTA, is not adversarially robust, and ECE on conformal-derived scores is strongly prevalence-sensitive.

The remainder of this paper is organized as follows. Section 2 positions the work against few-shot detectors, calibration, and conformal anomaly detection. Section 3 presents the decoupled pipeline, the LOIO conformal route, the attainable-alpha analysis, and SC3R. Section 4 describes datasets, protocols, and claim rules. Section 5 reports calibration comparisons, uniformity tests, operational false-alarm behavior, SC3R evaluations with their pre-specified gate, and diagnostics. Section 6 discusses limitations, and Section 7 concludes.

2. Related Work

2.1. Few-Shot Industrial Anomaly Detection

Classical industrial AD often assumes normal-only training data and detects deviations at test time. Patch distribution, self-supervised, and reconstruction-discriminative families such as PaDiM, CutPaste, and DRAEM provide important context for this literature [7–9]. PatchCore [2] made memory-bank patch retrieval a strong baseline by storing representative normal features and scoring test patches by nearest-neighbor distance. AnomalyDINO [3] updates this paradigm for few-shot inspection with frozen DINOv2 representations. The few-shot AD literature has also produced registration-based adaptation (RegAD [10]), graph-structured memory (GraphCore [11]), fast feature reconstruction (FastRecon [12]), vision–language zero-/few-shot scoring (WinCLIP [13]; CLIP–SAM collaboration [14] and cross-modal prompt regularization [15] continue this line), and in-context residual learning (InCTRL [16]). These methods are strong rankers, but the reference state can remain nontrivial at inference time and the raw score scale is not a calibrated probability; calibration under shift is also known to degrade for deep models generally [17]. Our work is complementary: it does not compete on ranking but asks how any such few-shot detector should expose decision reliability.

2.2. Frozen-Feature Subspace Detectors

Subspace approaches replace explicit patch retrieval with a compact normal subspace. SubspaceAD [4] shows that frozen DINOv2 patch features and PCA/subspace residuals are already strong for few-shot anomaly ranking; the ranking mechanism is therefore settled prior art. Our contribution starts after the residual is computed: how should a low-storage subspace detector expose reliability under low-shot and shift, and what can its alarms guarantee?

2.3. Calibration and Robustness in Few-Shot AD

Calibration asks whether predicted confidence agrees with empirical correctness [18, 19]. Standard post-hoc calibrators include Platt scaling [18], temperature scaling [19], histogram binning [20], and isotonic regression [21]; we compare against all of them under a shared label-free protocol. In anomaly detection this is subtle because labels are sparse, anomaly types are open-ended, and the score distribution changes across classes. Reliability analyses of DINOv2-based few-shot AD report poor ECE and strong FGSM fragility [5]. We therefore evaluate ECE, Brier score, and NLL separately from AUROC/AP, and report adversarial results, including FGSM-style perturbations [22], as diagnostics rather than robustness improvements.

2.4. Conformal Prediction and Reliability Routing

Conformal prediction converts nonconformity scores into p-values under exchangeability assumptions [23–25]. This is attractive for normal-only few-shot AD because support normal images can define a reference distribution without requiring real anomalies. Conformal anomaly detection itself is established: leave-one-out, bootstrap, and cross-conformal anomaly detectors have been analyzed directly [6], weighted conformal detection in low-data regimes with resolution and effective-sample-size analyses exists [26], general nonconformity toolkits are available [27], and few-shot conformal prediction with auxiliary tasks addresses the small-calibration-set problem [28]. Conformal AD, LOIO calibration, and weighted conformal p-values are thus established mechanisms, and our contribution begins where they stop in the few-shot regime: at the resolution floor $1/(k+1)$ that no target-only k-shot conformal alarm can cross. We prove the floor algebraically, audit corruption-shift deviations empirically, and introduce source-validated cross-category thresholding that operates below the floor without target anomaly labels. Gated-expert models, including SAGE-style shared/specialized routing [29], motivate the broader idea of choosing conservative or shift-specialized reliability paths. CRR uses this routing lens, but its main reported route is intentionally simple: fixed LOIO conformal reliability. Learned or weighted gates are treated as ablations unless they improve reliability without test-label expert selection.

3. Method

3.1. Problem Setup

For each category c, we observe a support set $S_c = \{x_i\}_{i=1}^{k}$ containing only normal images. The test set contains normal and anomalous images, but anomaly labels are used only for evaluation, not for fitting the main detector or calibrator. The goal is to produce: (i) a raw anomaly ranking score $s(x)$, (ii) a probability-like reliability score $\hat{p}(y = 1 \mid x)$ or conformal anomaly confidence, and (iii) optional pixel-level maps. Figure 1 gives an overview: a frozen backbone feeds a ranking lane that is never modified, a label-free reliability lane whose conformal alarms hit the attainable-alpha floor, and the SC3R lane that crosses the floor with source-class normals.

3.2. Frozen Patch Features and Subspace Ranking

We extract patch features with a frozen DINOv2 ViT-S/14 backbone. Let $z_{ij} \in \mathbb{R}^d$ denote patch j from image i. A PCA/subspace model is fit on support patches. With support mean μ and retained principal directions $U \in \mathbb{R}^{d\times m}$, the patch residual is

$$r(z) = \left\|(z-\mu) - UU^\top(z-\mu)\right\|_2. \tag{1}$$

Patch residuals form an anomaly map. The image-level ranking score is a high-residual aggregation,

$$s(x) = \mathrm{Agg}_j \; r(z_j), \tag{2}$$

where Agg is the patch maximum in the clean accuracy–storage benchmark and the mean of the top-1% patch residuals (ρ=0.01) in the conformal pipeline; the top-fraction variant is a smoothed maximum that stabilizes the LOIO calibration scores. We state this explicitly because the two protocols use different aggregators, and Section 5 reports an aggregation-sensitivity ablation. Importantly, this raw score is the quantity used for AUROC and AP. Calibration routes do not change the ranking score.

3.3. Vector and Shift-Aware Calibration

The first reliability family is post-hoc logistic calibration. A scalar Platt scaler maps $s(x)$ to a probability, while Vector Platt uses a richer descriptor

$$\phi(x) = \left[s_{\mathrm{pca}}(x),\; s_{\mathrm{head}}(x),\; |\tilde{s}_{\mathrm{pca}}(x) - \tilde{s}_{\mathrm{head}}(x)|\right], \tag{3}$$

where s_{head} is a lightweight synthetic-anomaly head score and tildes denote normalized scores. The head is a small MLP (384→256→1, dropout 0.1) trained with binary cross-entropy to separate support normal patches from synthetically perturbed patches; a patch perturbation resamples a support patch and adds Gaussian noise plus a sign shift in feature space. Its image-level score is the maximum patch probability. The head participates only in the calibration descriptor $\phi(x)$, never in the anomaly ranking. The calibrated probability is

$$\hat{p}_\theta(x) = \sigma(w^\top \phi(x) + b). \tag{4}$$

Shift-Aware Platt augments $\phi(x)$ with five shift descriptors: the maximum and mean distance of test patches to the support feature center (scaled by the support 95th-percentile norm), the mean and standard deviation of patch residual scores, and a residual concentration ratio (max over mean). Weighted Platt reweights calibration samples by density-ratio estimates from a logistic domain classifier, following weighted conformal prediction under covariate shift [30]. These variants are useful for structured shifts but can over-adapt, motivating a conservative conformal route.

Calibrator training protocol and label source. All calibrators in the main protocol are trained without any real anomaly labels (mode `normal_synthetic`). The calibration set consists of the k support normal images (label 0) and an equal number of synthetic images (label 1) obtained by replacing a random 25% of a support image's patch features with perturbed support patches as above. Vector-family calibrators fit a multivariate logistic model on $\phi(x)$ by gradient descent with L_2 regularization; the memory-bank and plain-subspace baselines, which expose no calibration descriptor, use scalar Platt on the raw score under the same synthetic protocol. The standard scalar calibrators evaluated in Section 5—temperature scaling [19], isotonic regression [21], and histogram binning [20]—are fit per class on exactly the same calibration set, mapping the raw score $s(x)$ to a probability; since raw residuals are not logits, temperature scaling is applied after centering at the midpoint of the two calibration class means, $\hat{p}(x) = \sigma((s(x)-c)/T)$, with only T

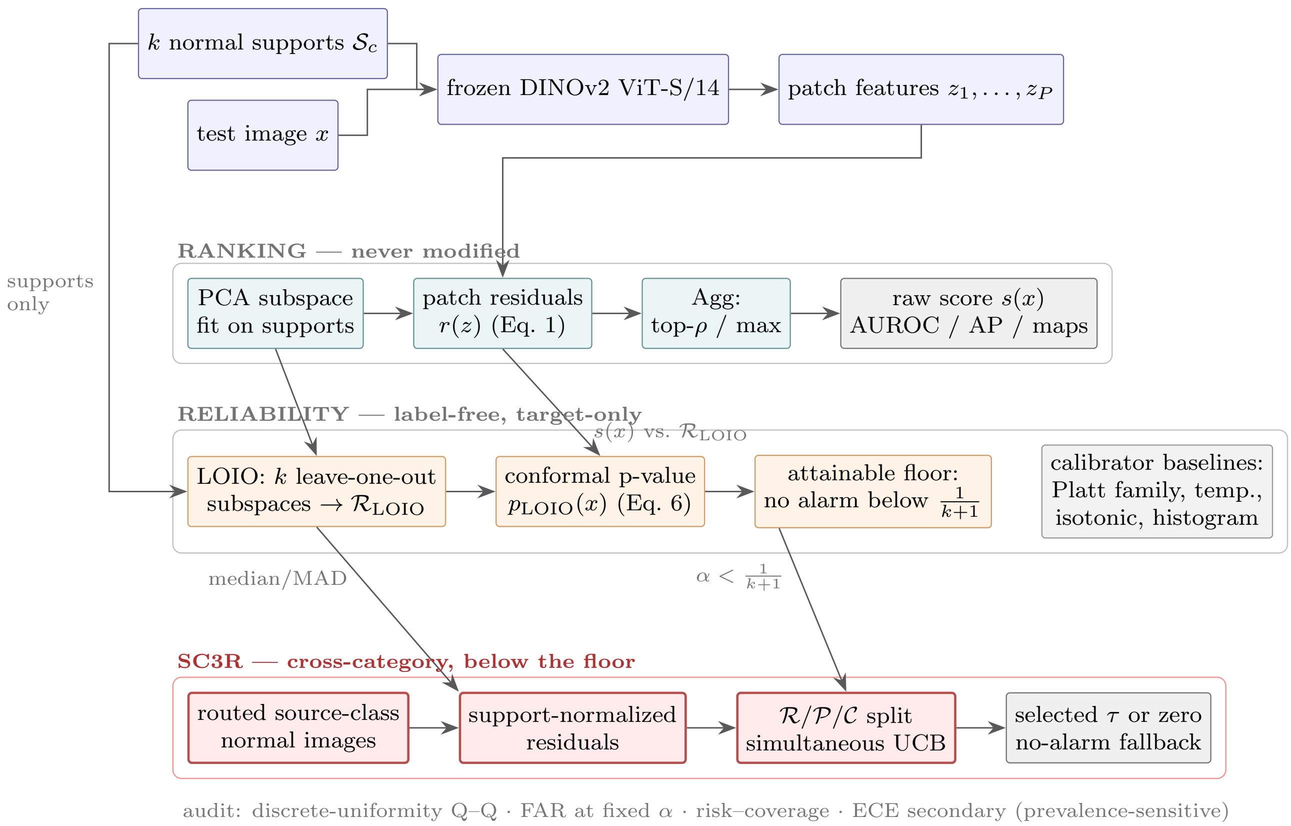

Figure 1: CRR overview. A frozen DINOv2 backbone feeds two decoupled paths: the *ranking* lane (PCA residuals, never modified; used for AUROC/AP and localization) and the label-free *reliability* lane (LOIO conformal p-values from the k supports, with Platt-family and standard scalar calibrators as baselines). Target-only alarms are structurally silent below $1/(k+1)$. The strict SC3R lane pools routed source-class normals, separates reference, proposal, and certification classes, and selects a threshold only when its simultaneous source-domain upper bound passes; otherwise it returns a no-alarm threshold.

learned. An `anomaly_val` mode that calibrates on 10% held-out real anomalies is reported only as an explicitly labeled upper bound. This symmetric, label-free protocol is what makes the LOIO-versus-calibrator comparison fair: every family sees exactly the same k support normals and no test information.

3.4. LOIO Conformal Reliability

For $k \geq 2$, CRR constructs normal calibration residuals by leave-one-image-out support splitting. For each support image x_i, a subspace is fit on $\mathcal{S}_c \setminus \{x_i\}$ and the held-out image receives a residual score $s_{-i}(x_i)$. The resulting set

$$\mathcal{R}_{\text{LOIO}} = \{s_{-i}(x_i) : x_i \in \mathcal{S}_c\} \tag{5}$$

serves as normal nonconformity evidence. For a test image x, the conformal p-value is

$$p_{\text{LOIO}}(x) = \frac{1 + \sum_{r \in \mathcal{R}_{\text{LOIO}}} \mathbf{1}\{r \geq s(x)\}}{1 + |\mathcal{R}_{\text{LOIO}}|}. \tag{6}$$

We report $p_{\text{conf}}(x) = 1 - p_{\text{LOIO}}(x)$ as a probability-like anomaly reliability view. It should be interpreted as relative extremeness against held-out support normals, not as a fully supervised posterior probability. Weighted conformal replaces the uniform count by density-ratio weights [30] and reports effective sample size; empirically, it is less stable than LOIO on the main benchmark.

Exchangeability caveat. The construction above is deliberately asymmetric: each calibration residual $s_{-i}(x_i)$ comes from a subspace fit on $k-1$ images, while the test score $s(x)$ uses all k. The two statistics are therefore not exactly exchangeable, and Eq. (6) is a leave-one-out approximation in the spirit of cross-conformal and jackknife+ inference [6, 23, 31] rather than a split-conformal p-value with a formal finite-sample guarantee [32, 33]. We handle this empirically on two fronts. First, a matched-LOIO audit scores held-out support and test images under the same fold-specific subspaces (pairing $s_{-i}(x)$ with r_{-i}, the jackknife+ pairing): it does not improve the operational false-alarm/power trade-off (Section 5), so the simpler construction is retained. Second, rather than assuming validity, we audit it directly: Section 5 reports discrete-grid diagnostics of the normal p-values. Their reference distribution is simulated under an idealized exchangeable null while preserving shared-calibration dependence within class–seed clusters; they localize empirical deviations but are not a proof of exchangeability.

3.5. Attainable Alpha and the Few-Shot Resolution Floor

Proposition 1 (attainable-alpha floor). For any finite calibration scores $r_1, \dots, r_k$, a p-value of the form in Eq. (6) belongs to $\{j/(k+1) : j = 1, \dots, k+1\}$. Hence the alarm $\mathbf{1}\{p(x) \le \alpha\}$ is identically zero for $\alpha < 1/(k+1)$.

Proof. The exceedance count $\sum_i \mathbf{1}\{r_i \ge s(x)\}$ is an integer between 0 and k. Adding one and dividing by $k+1$ gives the stated grid and lower bound. No exchangeability assumption is needed for this algebraic claim. □

At k=4 this floor is 0.2; at k=8 it is $1/9 \approx 0.111$. We make the resolution limit explicit with an attainable-alpha analysis that reports, for each nominal α, the nearest attainable grid point and the empirical false-alarm rate achieved there. This exact structural fact defines the regime in which cross-category support may be useful; it is not presented as the algorithmic novelty by itself. The textbook remedy for discreteness is a smoothed (randomized) conformal p-value, which uses an independent uniform tie-break and is uniform under the corresponding exchangeability assumptions. Section 5 evaluates it as a baseline: randomization crosses the floor but does not by itself address the observed support–test shift.

3.6. Source-Conditioned Cross-Category Reliability Routing (SC3R)

The attainable-alpha floor motivates the main routing contribution. In multi-category factories, normal images from *other* categories (source classes) are plentiful even when the target category has only k normal supports. SC3R uses them without target anomaly labels. First, target and source residuals are made comparable by support normalization: each raw score is centered and scaled by the robust median/MAD statistics of its own class's LOIO support residuals. This normalization is an empirical comparability device, not a proof that class distributions coincide. In matched-condition mode, source images share the target's corruption condition and condition metadata is therefore required. Clean-source always uses clean normals. The condition-agnostic ablation takes the median score across all available condition views of each base source image, so repeated corruptions are not counted as independent observations; deterministic mismatched-condition routing is only a negative control. Pooling source normals gives finer p-value resolution than the k target supports and permits thresholds below $1/(k+1)$.

3.7. Nested Source Certification and Transfer Boundary

For a fixed target, source classes are partitioned before certification into reference classes $\mathcal{R}$, proposal classes $\mathcal{P}$, and certification classes $\mathcal{C}$. Scores from $\mathcal{R}$ define conformal p-values. The distinct p-values on $\mathcal{P}$ propose at most M positive candidate thresholds. For candidate τ, certification unit i has loss $L_i(\tau) \in [0, 1]$. An image unit uses its Bernoulli alarm indicator and targets risk for the fixed source-image mixture. A class unit averages alarms within a held-out source class and targets a new draw from a source-category meta-population. Let A be the number of reported α levels. For the class unit with n classes, SC3R uses

$$U_{\mathrm{H}}(\tau) = \min\left\{ 1,\ \bar{L}(\tau) + \sqrt{\frac{\log(2AM/\delta)}{2n}} \right\}. \quad (7)$$

For independent Bernoulli image alarms, we use the exact one-sided binomial (Clopper–Pearson) upper bound with tail probability $\delta/(2AM)$. The factor $2A$ allocates the family error across both units and all reported levels; M allocates it across candidates. SC3R selects the largest proposed τ whose corresponding upper bound is at most α. If none passes, $\tau = 0$ gives a deterministic no-alarm fallback. Reference, proposal, and certification classes are disjoint; target test images and labels enter none of these stages.

Proposition 2 (conditional source certificate). Conditional on the reference and proposal stages, suppose certification units are independent draws from the declared source-unit population: Bernoulli losses for the image analysis, or bounded losses in $[0, 1]$ for the class analysis. Then, with probability at least $1 - \delta$, simultaneously across the A levels and both unit definitions, the corresponding source risk of every selected threshold is at most its α.

Proof. For a fixed level, unit, and candidate, the one-sided Clopper–Pearson construction or Hoeffding's inequality fails with probability at most $\delta/(2AM)$. A union bound over M candidates, A levels, and two units gives total failure probability at most δ; no independence between these bounds is required. Selection occurs only on this simultaneous event, so each passing candidate has risk at most its level. The zero fallback has zero alarm risk. □

Corollary 1 (conditional target transfer). On the event in Proposition 2, target risk is also at most α if, for every proposed threshold, the target normal alarm probability is no larger than the corresponding source-unit risk. The same follows if the target category is exchangeable with the certification-category meta-population.

The corollary's dominance or category-exchangeability condition is an assumption, not a consequence of robust normalization or matched routing. The two source certificates answer different questions: an image certificate for a fixed archive mixture cannot be relabeled a new-category guarantee, while a class certificate with few categories may be vacuous and cannot be replaced by the image result. The historical results in Section 5 predate this three-way split and remain labeled empirical until the frozen nested protocol is rerun. A direct pooled-source conformal baseline uses the entire routed source pool with threshold α and receives no artificial data restriction; unlike SC3R, it performs no proposal/certification step.

The broader routing component is inspired by the shared-versus-specialized principle in gated-expert systems. Candidate reliability views include Vector Platt, Shift-Aware Platt, weighted Platt, LOIO conformal, and weighted conformal. We evaluate fixed routes, validation-ECE selection, and simple no-label rules based on shift descriptors and effective sample size. Because learned gates did not dominate fixed

LOIO across all settings, the main paper reports fixed LOIO as the target-only reference and treats learned gates as diagnostic ablations; it is not called a safe route under distribution shift.

3.8. Selective Diagnostics

Beyond producing a calibrated alarm score, CRR records entropy, expert disagreement, conformal confidence, and effective sample size as selective-risk signals. These quantities are not used to change the anomaly ranking. Instead, they support deployment diagnostics such as abstention, risk-coverage analysis, and manual review of low-confidence alarms.

4. Experiments

4.1. Datasets and Few-Shot Splits

We evaluate on MVTec AD [34] and VisA [35]. MVTec AD contains industrial object and texture categories with pixel-level defect masks. VisA contains a broader set of industrial categories and is used as the main full-scale corruption and calibration-shift benchmark. For each class, support sets are sampled from normal training images with fixed seeds. Unless otherwise specified, $k \in \{1, 2, 4, 8\}$ is used for clean accuracy-storage studies, and $k \in \{4, 8\}$ is used for the full conformal corruption study because LOIO conformal requires at least two support images and is most relevant in moderate few-shot settings.

4.2. Evaluation Protocols

We organize experiments into five protocol groups. First, the clean few-shot protocol evaluates image AUROC, AP, calibration, and storage on MVTec and VisA. Second, the accuracy-storage protocol compares PCA64 and PCA128 to test whether a small subspace increase improves accuracy without approaching memory-bank storage. Third, the corruption reliability protocol evaluates Gaussian noise, blur, brightness/contrast shift, and JPEG compression at $k \in \{4, 8\}$ with at most 120 label-stratified images per case: on VisA across all 12 classes and five seeds (480 cases, 56,000 image rows), and on MVTec across all 15 classes and three seeds (360 cases, 37,464 image rows); VisA serves as the primary full-scale benchmark with the larger seed budget, while the MVTec replication uses three seeds for compute reasons, which its paired per-cell statistics account for. Fourth, representative routing protocols evaluate MVTec→VisA, VisA→MVTec, leave-one-class-out (LOCO), and within-dataset class splits. Fifth, the operational false-alarm protocol thresholds conformal p-values at nominal levels $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$ and reports false-alarm rate on normal images, detection power, alarm precision, and the attainable-alpha structure; the SC3R protocol additionally selects thresholds on held-out source-class normals with label-stratified random image sampling and a fixed sampling manifest, and is judged against a decision gate specified before the final stratified evaluation (nonzero power at $\alpha \leq 0.10$, mean false-alarm rate at most α+0.02, no-harm at least 80%, no target anomaly labels, hierarchical class–seed bootstrap support).

4.3. Naming Convention

CRR denotes the overall framework (decoupled ranking plus reliability layer). The detector instantiations are named by their subspace configuration: CRR/PCA64 (default, also called CalibSubspaceHead in configuration files) and CRR/PCA128. Reliability views are named by their route: scalar/Vector/Shift-Aware/weighted Platt, temperature scaling, isotonic regression, histogram binning, LOIO conformal, and weighted conformal. Tables use these names consistently.

4.4. Baselines and Ablations

We compare against a controlled nearest-neighbor memory bank using DINOv2, a SubspaceAD-style PCA residual baseline, scalar Platt, Vector Platt, Shift-Aware Platt, weighted Platt, temperature scaling, isotonic regression, histogram binning, LOIO conformal, and weighted conformal. The controlled memory-bank implementation is not an official reproduction of PatchCore or AnomalyDINO; it holds the frozen DINOv2 feature pipeline and data split fixed to isolate the scoring and storage trade-off. In addition, we run the official AnomalyDINO code (ViT-S/14 at 448px, agnostic preprocessing) on both MVTec and VisA (official 1-cls split) for $k \in \{1, 4, 8\}$ with three seeds and report it as a separate accuracy reference. As the vision–language representative we include WinCLIP [13], but only as reported numbers: no official implementation is released, and when we audited the leading community reimplementation under its own protocol (ViT-B-16+240, LAION-400M; zero-shot and 1-shot with its fixed support lists, three runs), it reproduced 71.7/77.4 mean image AUROC on MVTec and 66.0/70.0 on VisA against the reported 91.8/93.1 and 78.1/83.8—a roughly 20-point gap that the reimplementation's own authors also report and that we cannot attribute. This is precisely why our claim rules separate audited rows from reported rows, and WinCLIP therefore enters Table 1 as a cited reference, not an audited baseline. Official SubspaceAD representative runs are included as a novelty guardrail. Their strong performance confirms that frozen DINOv2 subspace residuals are an existing competitive ranking mechanism; the residual is inherited, and CRR's contribution begins at the reliability of the alarms built on it.

4.5. Metrics

We separate ranking metrics from reliability metrics. AUROC and AP are computed from the raw PCA/subspace residual score $s(x)$. Pixel-level localization is measured by pixel AUROC and AU-PRO when masks are available. Reliability is measured operationally by empirical false-alarm rate at fixed α, detection power, alarm precision, discrete uniformity of normal p-values, and risk–coverage curves;

expected calibration error (ECE), Brier score, and negative log-likelihood (NLL) are reported as secondary metrics using the selected reliability probability or probability-like conformal view. Storage is measured as retained reference/model state per class. Robustness experiments include corruption shift and a label-aware PCA-residual FGSM surrogate; the latter is used only as a fragility diagnostic.

4.6. Reproducibility and Claim Rules

The historical experiments were run on a single NVIDIA RTX 5090 GPU with PyTorch, with DINOv2 patch features cached in fp32. The final evidence package is required to archive the exact software versions and index every run by dataset, class, k, seed, model variant, calibration mode, and corruption/attack setting. The revised exporters write per-image predictions, base-image identities, support manifests, corruption parameters, and source-class partitions; an automated audit rejects duplicate cells, missing support statistics, and support–test overlap. A numerical result is treated as submission evidence only after these artifacts and their checksums are present. The core claim rule is that reliability routes cannot claim ranking improvement unless the raw score changes. Conversely, ranking methods cannot claim calibrated probabilities unless calibration metrics improve. This separation avoids conflating a better sorted list with a more trustworthy alarm probability.

5. Results

5.1. Clean Accuracy and Storage

Table 1 summarizes clean image-level performance and storage. The official AnomalyDINO reproduction (released code, 3 seeds) leads image AUROC at every k on both datasets: 0.965, 0.976, and 0.980 on MVTec; and 0.857, 0.912, and 0.926 on VisA, matching its published VisA numbers. We report this as an explicit accuracy reference: neither the controlled nearest-neighbor baseline nor CRR claims ranking superiority. CalibSubspaceHead is competitive with the controlled DINOv2 nearest-neighbor memory-bank baseline while being far more compact: PCA64 stores about 0.472 MB, compared with roughly 2–6 MB for the controlled memory bank. On VisA, the PCA128 ablation improves AUROC by 0.011–0.016 over PCA64 while keeping storage below 0.6 MB. Historical pointwise class–seed bootstrap intervals excluded zero at each k; their lower bounds ranged from 0.003 to 0.010. The controlled rows must not be read as official PatchCore or AnomalyDINO reproductions; only the official AnomalyDINO rows are. Two entries in Table 1 deserve explicit anatomy so they are not mistaken for placeholders or a strawman. Under the label-free scalar-Platt protocol, the controlled memory-bank baseline's probability collapses to a constant: at $k \leq 2$ every support patch self-matches (distance zero, probability $\approx$1), while at $k \geq 4$ the 4096-patch memory cap forces subsampling (probability $\approx$0). Its ECE therefore equals a prevalence complement (0.277/0.693) with Brier $\approx$ ECE—the signature of a constant probability—and its storage saturates at exactly 6.000 MB (4096 patches $\times$ 384 dims $\times$ 4 bytes). These values diagnose that raw nearest-neighbor distances resist naive label-free calibration; they are not competitive calibration results.

Storage accounting and compute cost. The per-class figures above cover only the class-specific reference state and deliberately exclude the shared frozen DINOv2 ViT-S/14 backbone ($\approx$22M parameters, $\approx$84 MB in fp32), which every compared method requires once per deployment. The 0.472 MB CRR state decomposes into the PCA subspace (64 components $\times$ 384 dims in fp32 $\approx$ 0.098 MB, matching the observed 0.094 MB PCA128–PCA64 increment) and the synthetic-anomaly head and calibrator ($\approx$0.37 MB); the controlled memory bank stores up to 4096 patch vectors ($\leq$6.0 MB). At inference, cached-feature scoring costs about 1.3 ms per image for the subspace detector versus 5–13 ms for the nearest-neighbor bank, while the shared backbone forward dominates end-to-end latency (order of seconds per batch on first pass, identical across methods). Fitting the k leave-one-out subspaces for LOIO adds a one-time per-class setup cost of k PCA fits on at most $k\times$1369 patch vectors, i.e., well under a second at $k \leq 8$.

Aggregation sensitivity. Because the clean benchmark aggregates patch residuals by max while the conformal pipeline uses the top-1% mean, Table 13 (Appendix A) quantifies sensitivity on representative classes. The observed spread is 0.01–0.02 AUROC: moderate top-ρ pooling ($\rho \in [0.005, 0.02]$) is slightly better than max on MVTec, while very wide pooling (ρ=0.05) hurts VisA. Both reported configurations lie within 0.01–0.02 of the best tested setting, suggesting that the representative ranking comparison is not driven by this choice; it does not establish invariance outside the tested grid.

5.2. Full VisA Conformal Reliability

Table 2 gives the main calibration-error result. Throughout the paper, ECE uses 15 equal-width bins; because few-shot conformal confidences occupy at most k+1 distinct values, we also report an equal-mass adaptive ECE below, and Section 5.4 replaces ECE with conformal-native diagnostics for the central claim. Across 56,000 full VisA corruption rows, LOIO conformal obtains pooled ECE 0.0766 while preserving the same PCA/subspace ranking. The strongest regime is $k = 4$, where pooled ECE drops to 0.0391. At $k = 8$, ECE increases to 0.1140 but remains better than Platt-style baselines in every corruption cell. The pooled reliability diagrams in Figure 2 visualize both effects: confidences occupy only the k+1 attainable levels, and the gap below the diagonal—overconfidence on normal images—widens from k=4 to k=8 and is larger on MVTec than on VisA. The discrete-grid analysis in Section 5.4 is consistent with clean-support/corrupted-test residual mismatch; the hypothesis that a tighter k=8 subspace amplifies it is not isolated causally.

Table 12 (Appendix A) isolates corruption effects. LOIO conformal improves over Vector Platt and Shift-Aware Platt for blur, brightness/contrast shift, Gaussian noise, and JPEG at both $k = 4$ and $k = 8$. Weighted conformal is a useful

Table 1
Clean image-level performance and storage (means over classes and 5 seeds; official rows: 3 seeds). Best accuracy within each dataset/k block is bold. Official AnomalyDINO rows are reproductions with the released code under its own protocol (ECE/storage not audited, left blank). WinCLIP rows are the numbers reported in [13]: no official implementation is released, and our audit of the leading community reimplementation falls about 20 AUROC points short of these numbers (see text), so WinCLIP is cited rather than audited. Controlled-NN ECE is diagnostic, not competitive (degenerate constant probability; see text), and CRR ECE at k=1 uses the vector-Platt view since LOIO requires $k \geq 2$.

Dataset	k	Method	AUROC ↑	AP ↑	ECE ↓	MB ↓
MVTec	1	Official AnomalyDINO	**0.965**	**0.982**	–	–
		WinCLIP (reported)	0.931	–	–	–
		Controlled DINOv2 NN	0.914	0.952	0.277	2.005
		CRR / CalibSubspace	0.904	0.951	**0.274**	**0.472**
	4	Official AnomalyDINO	**0.976**	**0.984**	–	–
		WinCLIP (reported)	0.952	–	–	–
		Controlled DINOv2 NN	0.942	0.967	0.693	6.000
		CRR / CalibSubspace	0.937	0.967	**0.214**	**0.472**
	8	Official AnomalyDINO	**0.980**	**0.990**	–	–
		Controlled DINOv2 NN	0.948	0.970	0.693	6.000
		CRR / CalibSubspace	0.945	**0.972**	**0.154**	**0.472**
VisA	1	Official AnomalyDINO	**0.857**	**0.866**	–	–
		WinCLIP (reported)	0.838	–	–	–
		Controlled DINOv2 NN	0.804	0.809	0.434	2.005
		CRR / PCA64	0.823	0.834	**0.429**	**0.472**
		CRR / PCA128	0.834	0.842	0.430	0.566
	4	Official AnomalyDINO	**0.912**	**0.918**	–	–
		WinCLIP (reported)	0.873	–	–	–
		Controlled DINOv2 NN	0.862	0.861	0.543	6.000
		CRR / PCA64	0.870	0.883	0.284	**0.472**
		CRR / PCA128	0.885	0.895	**0.278**	0.566
	8	Official AnomalyDINO	**0.926**	**0.929**	–	–
		Controlled DINOv2 NN	0.873	0.870	0.533	6.000
		CRR / PCA64	0.880	0.891	0.205	**0.472**
		CRR / PCA128	0.897	0.905	**0.170**	0.566

Table 2
Full VisA conformal reliability under corruptions. The PCA/subspace ranking score is fixed; only the reliability view changes. Best reliability values within each group are bold.

Group	Reliability view	N	AUROC	AP	ECE ↓	Brier ↓	NLL ↓
All	LOIO conformal	56000	0.823	0.846	**0.077**	**0.187**	**0.686**
	Weighted conformal				0.166	0.244	0.813
k=4	LOIO conformal	28000	0.820	0.842	**0.039**	**0.187**	**0.760**
	Weighted conformal				0.216	0.259	0.925
k=8	LOIO conformal	28000	0.828	0.851	**0.114**	**0.188**	**0.612**
	Weighted conformal				0.116	0.229	0.701

ablation but not a safe default: it is competitive for selected $k = 8$ structured corruptions, yet it is much worse at $k = 4$ and under Gaussian noise.

5.3. Full MVTec Conformal Reliability

The same conformal protocol was run on all 15 MVTec classes ($k \in \{4, 8\}$, seeds 0–2, four corruptions, 37,464 image rows). The VisA conclusion replicates: LOIO conformal reliability attains an overall ECE of 0.0684 (0.0600 at k=4, 0.0769 at k=8) while preserving the raw ranking (image AUROC 0.912), and it improves ECE over both Vector Platt and Shift-Aware Platt in every one of the eight k–corruption cells, often by a factor of two to four (e.g., k=4 Gaussian noise: 0.033 versus 0.233 and 0.270; Figure 3). Weighted conformal is again markedly worse (ECE 0.31–0.45). Conformal reliability is therefore not a VisA-specific effect; it holds across both benchmarks under identical protocols.

5.4. Statistical Diagnostics and Standard Calibrators

Conformal p-values satisfy $P(p \leq \alpha) \leq \alpha$ on normal data under the relevant exchangeability conditions—a false-alarm statement, not a posterior-probability interpretation. We therefore inspect this property directly instead of relying on ECE. Because a k-shot LOIO p-value is discrete on the grid $\{j/(k+1)\}$, a continuous Kolmogorov–Smirnov test

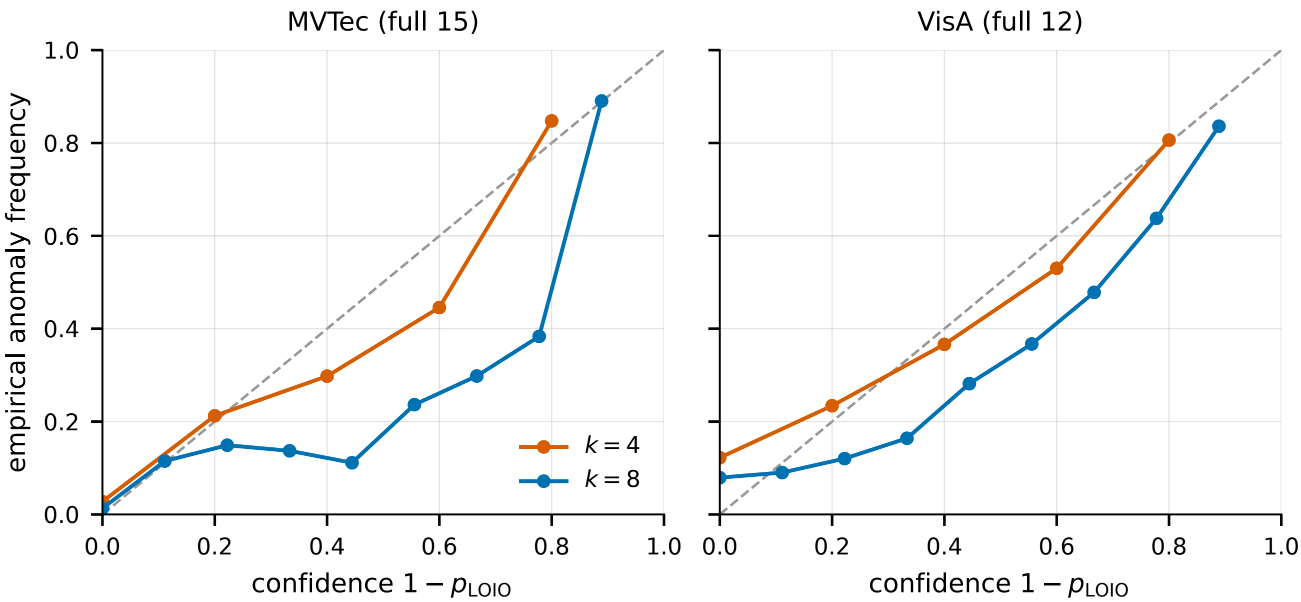

Figure 2: Reliability diagrams for LOIO conformal reliability on the full MVTec (left) and VisA (right) corruption benchmarks, pooled over classes, seeds, and corruptions. Points occupy the $k+1$ attainable confidence levels; gaps below the diagonal indicate overconfidence, which is larger at k=8, consistent with the uniformity analysis in Section 5.4.

against $U(0,1)$ is invalid by construction. Figure 4 instead compares the empirical CDF of normal p-values with the ideal discrete-uniform reference at every grid point. The descriptive deviations are conservative on VisA at k=4 (worst gap -0.10 at the 0.4 grid point), whereas VisA at k=8 and MVTec at both k become anti-conservative under corruption; for Gaussian noise on MVTec k=4, $\hat{F}(0.2) = 0.46$ against the reference value 0.20. Because rows share calibration sets, classes, and repeated seeds, we do not interpret a pooled iid test as confirmatory evidence. Our analysis code therefore uses an idealized exchangeable Monte Carlo reference that preserves shared-calibration dependence within class–seed clusters; category-level sensitivity remains part of the final artifact audit. The observed direction is consistent with corrupted normal residuals increasing relative to clean LOIO calibration scores, and the stronger k=8 deviation suggests that the tighter subspace may amplify this mismatch; neither observation isolates a causal mechanism. A matched-LOIO audit using fold-specific subspaces raises clean false alarms from 0.17 to 0.25 on representative MVTec k=4 with essentially unchanged power. This indicates that fold asymmetry alone does not explain the observed trade-off and motivates evaluating matched-condition source routing, without proving that condition mismatch is the sole cause.

Table 3 supplements point estimates with paired class–seed–corruption comparisons for LOIO against Vector Platt, Shift-Aware Platt, temperature scaling, isotonic regression, and histogram binning, all fit on the same label-free calibration set. The effect sizes favor LOIO in most settings (e.g., MVTec k=4: LOIO 0.120 ± 0.054 versus isotonic 0.222 ± 0.088, temperature 0.240 ± 0.104, and histogram binning 0.229 ± 0.099), but the margin narrows at k=8: against isotonic regression, LOIO is better in only 55% of MVTec cells and 62% of VisA cells. Against Shift-Aware Platt on VisA k=8, the mean difference is only -0.011 and LOIO is better in 51% of cells. The table retains the historical unadjusted Wilcoxon values as diagnostics; the released analysis code now adds a family-wise Holm correction, which must be regenerated from the per-cell artifacts before confirmatory significance language is used. Replacing equal-width with equal-mass (adaptive) ECE increases LOIO's per-cell ECE (MVTec: 0.120 → 0.160 at k=4, 0.135 → 0.182 at k=8; VisA: 0.132 → 0.227, 0.158 → 0.236) while preserving the observed ordering in the historical tables. Binning inflates the apparent margin, which is one more reason the operational analyses below, not ECE, carry the central claim.

5.5. Attainable Alpha and Operational False-Alarm Behavior

Calibration error alone does not answer the operator's question: "if I alarm at $p \leq \alpha$, how often is a good part flagged?" Table 4 reports the attainable-alpha analysis. The pattern is structural, not empirical: at k=4 every nominal level below 0.2 raises zero alarms on both datasets because no attainable p-value exists below the floor $1/(k+1)$, and at k=8 the same holds below $1/9$. Reporting these zeros as "conservative coverage" would be misleading; they are resolution artifacts of the few-shot calibration set.

Table 5 evaluates the first attainable operating point, α=0.20, on both full benchmarks. On VisA the LOIO alarm rule is conservative but functional (VisA k=4: false-alarm 0.14–0.16 with detection 0.58–0.61 and precision around 0.81); on full MVTec the same rule is anti-conservative under corruption at k=4, which the SC3R analysis below addresses directly. We note a practical pitfall uncovered during this analysis: p-values stored in single precision can exceed

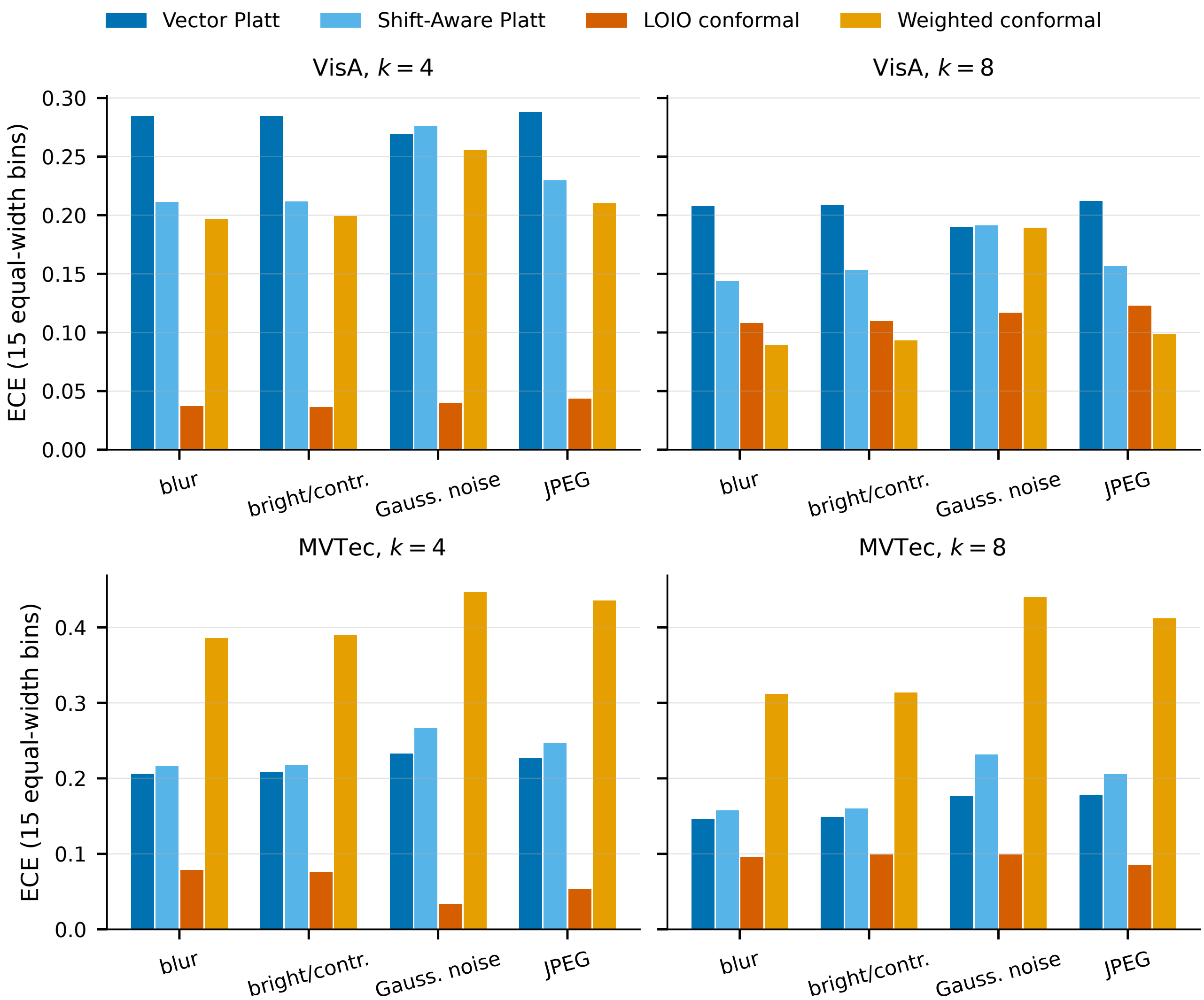

Figure 3: ECE by corruption and reliability view on full VisA (top) and full MVTec (bottom) at k=4 and k=8. LOIO conformal is the most stable view; weighted conformal is competitive only in selected VisA k=8 structured corruptions.

their exact rational value (e.g., $1/5 \to 0.20000000298$), so an exact comparison $p \leq \alpha$ silently discards every alarm at the floor. All thresholding in this paper uses a small tolerance, and we recommend the same in deployments.

5.6. SC3R: Operating Below the Target-Only Alpha Floor

Table 6 reports the historical SC3R analysis on all 15 label-stratified MVTec classes (k=4, seeds 0–2, five conditions, 675 evaluation cells). The decision gate required nonzero power below the target-only floor, mean false-alarm rate at most α+0.02, no-harm in at least 80% of class–seed–corruption cells, no target anomaly labels, and class–seed bootstrap support. In this analysis, matched-condition SC3R met the empirical gate: mean power was 0.216 and 0.416 at α=0.05 and 0.10, where the anchor was structurally silent. Mean false-alarm rates were 0.050, 0.105, and 0.216 for α=0.05, 0.10, and 0.20, respectively. Alarm precision exceeded 0.90 and no-harm rates were 89%/82%/89%. The reported pointwise 95% hierarchical intervals excluded zero in each evaluated sub-floor condition, but they do not provide simultaneous coverage across the family. Moreover, candidate selection and source assessment reused the same historical source evidence; these numbers are therefore empirical diagnostics, not independent certification. At α=0.20, the anchor attained higher power but false-alarm rates reached 0.27–0.46 under corruption, whereas SC3R stayed within 0.005–0.035 of nominal. Material exceptions remain: JPEG at α=0.10 (0.121) and Gaussian noise/JPEG at α=0.20 (0.226/0.235) exceeded the per-condition α+0.02 budget even though pooled means passed.

Replication at k=8. Table 7 repeats the historical protocol at k=8 (floor $1/9 \approx 0.111$), where the anchor is silent at $\alpha \in \{0.05, 0.10\}$. Matched-condition SC3R reaches mean power 0.547/0.693 at α=0.05/0.10 with precision

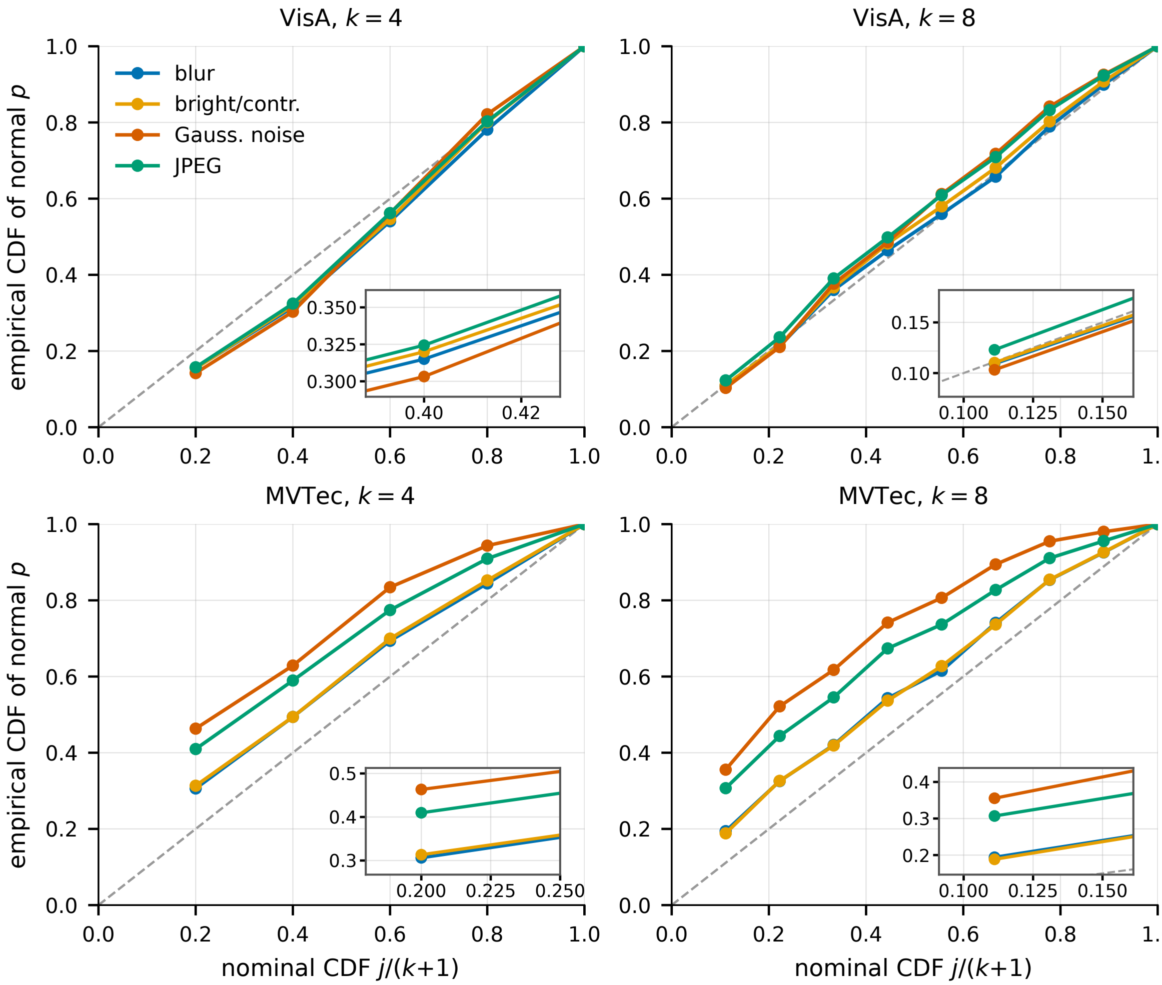

Figure 4: Discrete Q-Q analysis of normal-image LOIO p-values: empirical CDF at each attainable grid point $j/(k+1)$ against the idealized exchangeable reference (dashed diagonal), per corruption. Points above the diagonal indicate excess false alarms relative to that reference. VisA k=4 is conservative; VisA k=8 and MVTec drift anti-conservative under corruption, worst for Gaussian noise. The lower-right insets enlarge the first informative grid region while occupying data-free areas of their parent panels.

above 0.91; the reported per-condition intervals are again pointwise rather than simultaneous. False-alarm estimates are near nominal on clean and structured corruptions, but the empirical budget passes only at α=0.05: pooled rates are 0.052/0.121/0.230 and no-harm rates 80%/74%/81%. At α=0.10, false-alarm rates rise to 0.138 for Gaussian noise and 0.141 for JPEG. The hypothesis that these transformations induce more heterogeneous normal residual shifts is plausible but not causally isolated. Thus the historical evidence supports sub-floor operation, while heavy-tailed corruptions and the non-independent threshold analysis limit the strength of a control claim.

Why source validation rather than randomization. The standard remedy for the attainable-alpha floor is the smoothed conformal p-value, which breaks the k+1-point discreteness with an independent uniform tie-break and admits alarms at any nominal level. Table 8 evaluates this baseline on the same target-only LOIO residuals and test images as SC3R, reporting exact expectations over the randomization. Randomization does cross the floor, and on VisA—where the target-only rule is conservative—it behaves reasonably (FAR below nominal at reduced power). But it has no mechanism to correct the support–test mismatch: on MVTec its false-alarm rate exceeds the budget *even under the clean condition* (0.067 at α=0.05, reflecting the leave-one-out asymmetry) and reaches 2.3× nominal under Gaussian noise (0.115), inheriting the anti-conservative bias localized by the uniformity tests. SC3R attains comparable sub-floor power (0.216 vs. 0.216 at α=0.05; 0.416 vs. 0.431 at α=0.10) while producing near-nominal pooled false-alarm estimates on both evaluated datasets. Randomization addresses resolution; source validation additionally mitigates the observed source–test mismatch, without providing an unconditional target-domain guarantee.

Replication on VisA and cross-dataset source archives. Table 9 reports the same historical analysis on all 12 VisA classes (k=4, seeds 0–2). With within-VisA sources,

Table 3
LOIO conformal reliability versus Platt-family and standard scalar calibrators on the full corruption benchmarks. All calibrators are label-free and fit on the identical calibration set (k support normals plus k synthetic anomalies). Entries are means±std of per-cell (class×seed×corruption) ECE with 15 equal-width bins; Δ is LOIO minus baseline (negative favors LOIO), and the last column is the fraction of cells where LOIO is better. The displayed paired two-sided Wilcoxon p-values are historical unadjusted diagnostics; effect sizes, win fractions, and operational metrics carry the interpretation until the artifact-complete Holm-adjusted analysis is regenerated. Bold marks the lower per-cell mean.

dataset / k	baseline	base ECE	LOIO ECE	Δ	Wilcoxon p	LOIO better
MVTec k=4	Vector Platt	0.223±0.095	**0.120±0.054**	-0.104	2×10^{-23}	84%
	Shift-Aware Platt	0.241±0.104	**0.120±0.054**	-0.121	9×10^{-25}	88%
	Temperature scaling	0.240±0.104	**0.120±0.054**	-0.120	1×10^{-25}	87%
	Isotonic regression	0.222±0.088	**0.120±0.054**	-0.103	1×10^{-24}	87%
	Histogram binning	0.229±0.099	**0.120±0.054**	-0.110	1×10^{-23}	84%
	Scalar Platt	0.268±0.127	**0.120±0.054**	-0.148	3×10^{-29}	93%
MVTec k=8	Vector Platt	0.167±0.065	**0.135±0.055**	-0.033	5×10^{-9}	66%
	Shift-Aware Platt	0.192±0.088	**0.135±0.055**	-0.057	9×10^{-15}	75%
	Temperature scaling	0.177±0.080	**0.135±0.055**	-0.042	5×10^{-10}	69%
	Isotonic regression	0.157±0.069	**0.135±0.055**	-0.023	2×10^{-4}	55%
	Histogram binning	0.162±0.080	**0.135±0.055**	-0.027	4×10^{-5}	61%
	Scalar Platt	0.256±0.132	**0.135±0.055**	-0.121	2×10^{-27}	91%
VisA k=4	Vector Platt	0.286±0.060	**0.131±0.063**	-0.155	6×10^{-24}	97%
	Shift-Aware Platt	0.237±0.087	**0.131±0.063**	-0.106	3×10^{-19}	90%
	Temperature scaling	0.312±0.085	**0.132±0.059**	-0.180	1×10^{-40}	98%
	Isotonic regression	0.239±0.073	**0.132±0.059**	-0.106	4×10^{-33}	90%
	Histogram binning	0.301±0.087	**0.132±0.059**	-0.168	4×10^{-40}	96%
	Scalar Platt	0.304±0.145	**0.132±0.059**	-0.171	4×10^{-35}	90%
VisA k=8	Vector Platt	0.199±0.063	**0.151±0.054**	-0.048	3×10^{-14}	81%
	Shift-Aware Platt	0.162±0.075	**0.151±0.054**	-0.011	0.20	51%
	Temperature scaling	0.247±0.089	**0.158±0.051**	-0.089	7×10^{-23}	75%
	Isotonic regression	0.194±0.068	**0.158±0.051**	-0.035	5×10^{-8}	62%
	Histogram binning	0.242±0.097	**0.158±0.051**	-0.083	2×10^{-18}	69%
	Scalar Platt	0.418±0.100	**0.158±0.051**	-0.259	1×10^{-40}	98%

Table 4
Attainable-alpha structure of few-shot LOIO conformal p-values. With k support scores the smallest attainable p-value is $1/(k+1)$, so nominal levels below this floor structurally raise no alarms; empirical false-alarm rates (FAR, mean over classes/seeds/corruptions) then sit at the nearest attainable grid point rather than at the nominal level.

dataset	k	nominal α	floor	attainable	FAR	detection
VisA (full)	4	0.01	0.200	0.000	0.000	0.000
		0.05		0.000	0.000	0.000
		0.10		0.000	0.000	0.000
		0.20		0.200	0.152	0.599
	8	0.01	0.111	0.000	0.000	0.000
		0.05		0.000	0.000	0.000
		0.10		0.000	0.000	0.000
		0.20		0.111	0.111	0.537
MVTec (full)	4	0.01	0.200	0.000	0.000	0.000
		0.05		0.000	0.000	0.000
		0.10		0.000	0.000	0.000
		0.20		0.200	0.411	0.887
	8	0.01	0.111	0.000	0.000	0.000
		0.05		0.000	0.000	0.000
		0.10		0.000	0.000	0.000
		0.20		0.111	0.304	0.907

observed false-alarm rates were 0.051/0.095/0.192, sub-floor power was 0.144/0.351, precision exceeded 0.84, and no-harm was 89%/84%. The reported power-gain intervals excluded zero by condition but were pointwise. At α=0.20, no-harm fell to 78%. Replacing the source pool with 15 MVTec classes yielded observed false-alarm rates 0.013/0.040/0.098 and lower power 0.091/0.201/0.415. This is one directional stress test: it suggests a power–false-alarm trade-off for MVTec-to-VisA transfer, not a general fail-safe property of foreign archives.

Table 5
Operational false-alarm behavior of target-only LOIO conformal p-values at the first attainable operating point (α=0.20; alarms fire at $p \leq \alpha$ with a float tolerance). Rates are pooled over classes and seeds. On VisA the rule is conservative (FAR below nominal); on full MVTec it is anti-conservative under corruption, especially at k=4, motivating source-validated thresholding (Table 6).

dataset	corruption	k	FAR	detection	precision
VisA (full)	blur	4	0.155	0.610	0.806
		8	0.109	0.546	0.841
	bright/contr.	4	0.155	0.595	0.803
		8	0.110	0.533	0.836
	Gauss. noise	4	0.142	0.585	0.813
		8	0.103	0.522	0.843
	JPEG	4	0.157	0.604	0.803
		8	0.123	0.547	0.825
MVTec (full)	blur	4	0.306	0.887	0.872
		8	0.194	0.908	0.916
	bright/contr.	4	0.313	0.887	0.869
		8	0.188	0.908	0.919
	Gauss. noise	4	0.463	0.879	0.816
		8	0.355	0.898	0.855
	JPEG	4	0.410	0.890	0.836
		8	0.307	0.914	0.875

Table 6
Historical source-validated thresholding (SC3R, matched-condition source mode) versus the target-only LOIO anchor on all 15 label-stratified MVTec classes (k=4, seeds 0–2). Below $1/(k+1)$=0.2, the anchor cannot raise alarms; source pooling unlocks these operating points with false-alarm estimates near the pre-specified budget on the evaluated targets. Threshold selection and source assessment are not independent in this historical analysis. Bold marks cells satisfying FAR $\leq \alpha$+0.02 with nonzero power.

		SC3R (source-validated)			Target-only anchor		
corruption	α	FAR	power	prec.	FAR	power	prec.
clean	0.05	**0.050**	**0.267**	0.928	0.000	0.000	–
	0.10	**0.100**	**0.470**	0.957	0.000	0.000	–
	0.20	**0.209**	**0.768**	0.914	0.267	0.863	0.898
blur	0.05	**0.048**	**0.242**	0.915	0.000	0.000	–
	0.10	**0.099**	**0.468**	0.950	0.000	0.000	–
	0.20	**0.207**	**0.760**	0.904	0.269	0.860	0.895
bright/contr.	0.05	**0.050**	**0.254**	0.914	0.000	0.000	–
	0.10	**0.101**	**0.467**	0.954	0.000	0.000	–
	0.20	**0.205**	**0.769**	0.905	0.268	0.863	0.898
Gauss. noise	0.05	**0.048**	**0.184**	0.934	0.000	0.000	–
	0.10	**0.105**	**0.295**	0.929	0.000	0.000	–
	0.20	0.226	0.495	0.914	0.460	0.860	0.844
JPEG	0.05	**0.051**	**0.134**	0.911	0.000	0.000	–
	0.10	0.121	0.381	0.936	0.000	0.000	–
	0.20	0.235	0.633	0.917	0.434	0.867	0.854

5.7. ECE Is Prevalence-Sensitive for Conformal Scores

A further protocol warning emerged from prevalence stress tests: ECE computed on $1 - p$ conformal scores changes drastically with the anomaly prevalence of the evaluation set even when the ranking is fixed. On full VisA, LOIO ECE moves from 0.404 at 1% prevalence to 0.149 at 50%, and the LOIO-versus-weighted ranking reverses across this range. ECE on anomaly-balanced test sets therefore cannot certify deployment reliability of conformal-derived scores. We report ECE as a secondary metric only and rely on false-alarm rate, power, precision, and risk–coverage for operational claims. The risk–coverage view (Figure 5) also quantifies the value of abstention: ranking images by predictive entropy and abstaining on the most uncertain 20% halves the remaining LOIO calibration error on full MVTec (0.068 to 0.030) and improves it moderately on VisA (0.077 to 0.075 at 80% coverage, 0.066 at 70%), so selective operation is a practical complement to fixed-α alarms.

Table 7

SC3R at k=8 (matched-condition source mode) versus the target-only LOIO anchor on all 15 label-stratified MVTec classes (seeds 0–2). The target-only floor is $1/9 \approx 0.111$, so $\alpha \in \{0.05, 0.10\}$ are below the anchor's resolution. Bold marks cells satisfying FAR $\leq \alpha$+0.02 with nonzero power.

		SC3R (source-validated)			Target-only anchor		
corruption	α	FAR	power	prec.	FAR	power	prec.
clean	0.05	**0.053**	**0.764**	0.976	0.000	0.000	–
	0.10	**0.110**	**0.833**	0.936	0.000	0.000	–
	0.20	**0.216**	**0.898**	0.917	**0.182**	**0.903**	0.928
blur	0.05	**0.050**	**0.751**	0.976	0.000	0.000	–
	0.10	**0.110**	**0.813**	0.957	0.000	0.000	–
	0.20	0.221	0.900	0.917	**0.199**	**0.904**	0.922
bright/contr.	0.05	**0.051**	**0.770**	0.976	0.000	0.000	–
	0.10	**0.107**	**0.831**	0.958	0.000	0.000	–
	0.20	0.223	0.895	0.916	**0.185**	**0.904**	0.926
Gauss. noise	0.05	**0.035**	**0.155**	0.962	0.000	0.000	–
	0.10	0.138	0.398	0.951	0.000	0.000	–
	0.20	0.232	0.749	0.917	0.392	0.904	0.866
JPEG	0.05	0.071	0.296	0.972	0.000	0.000	–
	0.10	0.141	0.588	0.959	0.000	0.000	–
	0.20	0.260	0.826	0.918	0.380	0.911	0.875

Table 8

Why source validation, not randomization: smoothed (randomized) conformal p-values versus SC3R below the target-only floor (k=4, all classes, seeds 0–2, pooled over five conditions; randomized rates are exact expectations over the randomization). Randomization crosses the floor but inherits the target-only rule's bias—anti-conservative on MVTec even under the clean condition (FAR 0.067 at α=0.05) and up to $2.3\times$ nominal under Gaussian noise—whereas SC3R attains comparable power with false-alarm rates at nominal. Bold marks FAR $\leq \alpha$+0.02.

		Randomized p-value		SC3R	
dataset	α	FAR	power	FAR	power
MVTec	0.05	0.085	0.216	**0.049**	0.216
	0.10	0.170	0.431	**0.105**	0.416
VisA	0.05	**0.036**	0.150	**0.051**	0.144
	0.10	**0.072**	0.300	**0.095**	0.351

5.8. Routing, Pixel Localization, and Fragility Diagnostics

Table 10 evaluates protocol-locked representative routing. Fixed LOIO conformal reduces ECE relative to Vector Platt without using validation labels to select an expert. This supports the central CRR claim: reliability routing can be useful even when the route is fixed and conservative. SAGE-style gating remains a promising design inspiration, but the current main evidence favors simple LOIO routing over learned or weighted expert selection.

Pixel-level results (Table 11) show that the low-storage subspace detector is also competitive for localization: it matches or exceeds the controlled memory bank on pixel AUROC at every k and on AU-PRO from k=4. Robustness

Table 9

Historical SC3R replication on all 12 label-stratified VisA classes (k=4, seeds 0–2, matched-condition mode, pooled over five conditions). The source pool is either the other VisA classes or all 15 MVTec classes. Pointwise hierarchical power-gain intervals exclude zero in the reported sub-floor cells but are not simultaneous across corruptions. In the evaluated MVTec-to-VisA direction, observed false alarms are below nominal at reduced power. Bold marks FAR $\leq \alpha$+0.02 with nonzero power.

source pool	α	FAR	power	prec.	no-harm
VisA (within)	0.05	**0.051**	**0.144**	0.847	89%
	0.10	**0.095**	**0.351**	0.843	84%
	0.20	**0.192**	**0.637**	0.810	78%
MVTec (cross-dataset)	0.05	**0.013**	**0.091**	0.936	97%
	0.10	**0.040**	**0.201**	0.881	93%
	0.20	**0.098**	**0.415**	0.852	93%

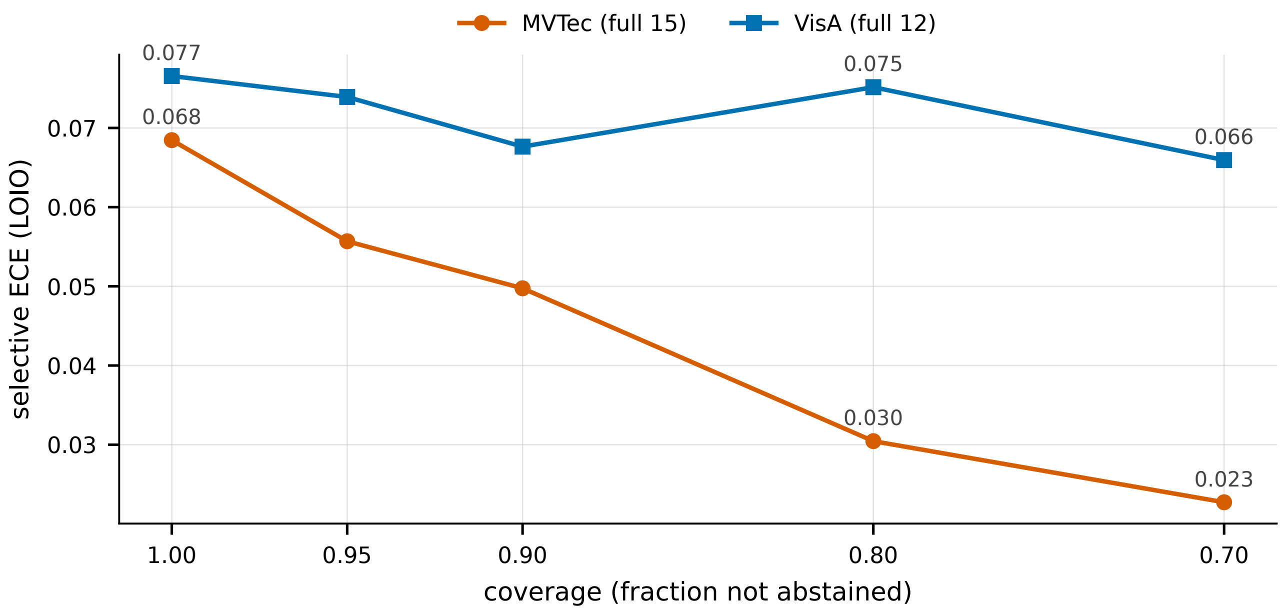

Figure 5: Overlaid risk–coverage curves for LOIO conformal reliability with entropy-ranked abstention on full MVTec and VisA. Abstaining on the 20% most uncertain images halves the remaining calibration error on MVTec.

Table 10
Protocol-locked representative routing. Fixed LOIO conformal does not require validation-label expert selection and reduces ECE relative to Vector Platt in every protocol; no-harm is the fraction of cells where the LOIO route does not degrade calibration.

Protocol	Vector ECE	LOIO ECE	Δ ECE	No-harm
MVTec→VisA	0.301	**0.099**	−0.203	100%
VisA→MVTec	0.304	**0.102**	−0.202	100%
LOCO	0.304	**0.116**	−0.188	100%
Within split	0.361	**0.154**	−0.207	100%

diagnostics are less favorable: the label-aware PCA-residual FGSM surrogate collapses image AUROC from 0.94 to 0.16 at ϵ=2/255—far below chance, i.e., the attack inverts the score ordering—and the drop shrinks rather than grows at larger ϵ (0.45 at 8/255), a non-monotonicity that signals objective sensitivity of the surrogate itself. These results are important negative evidence and motivate reliability diagnostics, but they do not support an adversarial robustness claim.

5.9. Novelty Guardrails

Table 14 (Appendix A) marks, for each ingredient, where prior work ends and this paper's claims begin. The official SubspaceAD representative run is especially important because it reaches high image and pixel AUROC in our protocol: the ranking mechanism is inherited, and the contributions of this paper start after it—at the attainable-alpha floor, its uniformity audit, and the source-validated thresholding that crosses it.

6. Limitations

CRR is a reliability framework, not a universal accuracy replacement or an MVTec AUROC state-of-the-art claim. Official SubspaceAD remains a strong ranking baseline; our contribution concerns compact calibration, operational diagnostics, and source-assisted routing.

CRR is not adversarially robust. Label-aware PCA-residual FGSM stress tests cause severe degradation and depend on the attack objective, so corruption robustness must not be extrapolated to adversarial robustness.

Conformal validity depends on representative normal support data. LOIO residuals and full-support test residuals are asymmetric, and the normal p-values lie on a finite grid. Section 5.4 therefore reports clustered, discrete-reference diagnostics rather than treating a pooled iid uniformity test as exact. Without the relevant exchangeability conditions, target false-alarm control is not guaranteed.

The reported SC3R tables are historical empirical evidence. Candidate selection and source assessment were not independent, and their bootstrap intervals were pointwise rather than simultaneous. The k=8 false-alarm budget passed only at α=0.05; at α=0.10 the pooled rate was 0.121

Table 11
Pixel-level localization and adversarial-fragility diagnostics on MVTec (means over classes and 5 seeds). Top: the low-storage subspace detector matches the controlled memory bank on pixel AUROC and AU-PRO. Bottom: a label-aware PCA-residual FGSM surrogate collapses image AUROC far below chance at small ε (score inversion), and the drop is non-monotonic in ε—evidence of objective sensitivity that we report as a diagnostic, not as a robustness claim.

Pixel localization

k	Method	pixel AUROC	AU-PRO
1	CRR / CalibSubspace	**0.944**	0.820
	Controlled DINOv2 NN	0.940	**0.845**
4	CRR / CalibSubspace	**0.957**	0.862
	Controlled DINOv2 NN	0.951	0.862
8	CRR / CalibSubspace	**0.960**	**0.876**
	Controlled DINOv2 NN	0.952	0.863

FGSM fragility diagnostic (image AUROC)

ε	k	clean → FGSM	rel. drop
2/255	4	0.937 → 0.158	83%
	8	0.945 → 0.175	82%
8/255	4	0.937 → 0.446	52%
	8	0.945 → 0.446	53%

against a 0.120 budget with 74% no-harm, concentrated in Gaussian noise and JPEG. VisA no-harm fell to 78% at α=0.20, and the single MVTec-to-VisA direction traded power for lower observed false alarms. Source certification alone does not imply unconditional target-domain control. SC3R also requires multiple source categories, while matched-condition routing assumes condition metadata or a separately evaluated condition detector.

Finally, ECE for conformal-derived scores is prevalence-sensitive. Calibration numbers on anomaly-balanced benchmarks should not be interpreted as deployment reliability; false-alarm rate, power, precision, and risk–coverage are the primary operational summaries.

7. Conclusion and future work

We introduced Conformal Reliability Routing, a decoupled framework for low-storage few-shot industrial anomaly detection. A target-only alarm built from k calibration scores cannot fire below $\alpha = 1/(k+1)$; the empirical audit further shows anti-conservative behavior under selected corruptions. SC3R addresses the resulting operating gap by pooling support-normalized source-class normals and selecting thresholds on held-out source classes. Under a prespecified evaluation gate, the historical SC3R analysis provides sub-floor detection power and near-nominal empirical false-alarm rates at k=4 on the evaluated MVTec targets. The pattern replicates on VisA, and the evaluated MVTec-to-VisA direction has lower observed false-alarm rates at reduced power. These are empirical transfer results rather than unconditional distribution-free target guarantees. The boundary is equally important: at k=8, Gaussian noise and JPEG erode the false-alarm budget above α=0.05. Supporting analyses show that the reliability layer leaves the inherited DINOv2 PCA/subspace ranking unchanged, improves secondary calibration metrics over standard scalar calibrators in most comparisons, and remains storage-light. CRR is therefore not an AUROC state-of-the-art or adversarial-robustness claim; it is a reliability framework whose current evidence identifies where source-assisted few-shot alarms appear useful, where they fail, and which source-to-target assumptions are required for stronger statements.

Future work is scoped by the boundaries above: SC3R with more seeds and per-condition source validation to close the Gaussian/JPEG budget gap, and an audited vision–language baseline, which currently blocks on the absence of an official WinCLIP implementation.

CRediT authorship contribution statement

CRediT authorship contribution statement

Given Family: Conceptualization, Methodology, Software, Validation, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets analyzed during the current study are publicly available: MVTec AD (`https://www.mvtec.com/company/research/datasets/mvtec-ad`) and VisA (`https://github.com/amazon-science/spot-diff`). The final review package will include derived per-image prediction tables, evaluation summaries, sampling and source-partition manifests, and checksums; the reviewer-accessible repository identifier must be inserted before submission.

Table 12
ECE comparison on the full VisA corruption benchmark. Lower is better. Bold indicates the best reliability view in each corruption cell; Figure 3 visualizes the same comparison for both benchmarks.

k	Corruption	Vector	Shift-aware	LOIO	Weighted
4	blur	0.284	0.211	**0.037**	0.197
	brightness/contrast	0.285	0.212	**0.036**	0.199
	gaussian noise	0.270	0.276	**0.040**	0.256
	jpeg	0.288	0.230	**0.043**	0.210
8	blur	0.208	0.144	0.108	**0.089**
	brightness/contrast	0.209	0.153	0.109	**0.093**
	gaussian noise	0.190	0.191	**0.117**	0.189
	jpeg	0.212	0.156	0.123	**0.099**

Table 13
Aggregation-sensitivity ablation: clean image AUROC (mean over representative classes, seeds 0–2) when patch residuals are pooled by max versus top-ρ mean. Bold marks the best aggregator per row; the max/top-1% choices used in the paper are within 0.01–0.02 AUROC of the per-row optimum.

dataset	k	max	top-0.5%	top-1%	top-2%	top-5%
MVTec	4	0.9571	0.9638	0.9667	**0.9693**	0.9621
	8	0.9578	0.9685	0.9740	**0.9759**	0.9727
VisA	4	0.9039	**0.9101**	0.9038	0.8944	0.8806
	8	0.9101	**0.9200**	0.9147	0.9071	0.8941

Code availability

The experimental code includes the conformal protocols, calibrators, corruption pipeline, artifact audits, and table/figure generators. A reviewer-accessible repository and immutable commit identifier must be inserted here before submission.

Declaration of generative AI and AI-assisted technologies

During manuscript preparation, the authors used OpenAI Codex for code review, test generation, statistical-claim auditing, and language editing. No experimental observation was generated or altered by the tool. The authors reviewed the resulting code and text and take responsibility for the content.

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A. Supplementary tables

This appendix collects supporting tables whose content is summarized in the main text: the per-corruption VisA ECE breakdown (Table 12; Figure 3 visualizes the same comparison), the aggregation-sensitivity ablation (Table 13), and the novelty-positioning summary (Table 14).

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Table 14
Novelty positioning: each ingredient has established prior art, and this paper's contribution begins after it. The right column marks where prior work ends and our claims start.

Work family	Established prior art	Our contribution starts after it
AnomalyDINO	DINOv2 few-shot patch/memory-bank AD	Reliability of the alarms such detectors raise
SubspaceAD	DINOv2 PCA/subspace residual ranking	Everything downstream of the residual
Calibration/robustness studies	ECE, Platt scaling, FGSM fragility	Conformal-native audit; full calibrator toolbox under one label-free protocol
Conformal AD	P-values from nonconformity scores	The few-shot floor $1/(k+1)$ and source-validated thresholding below it
SAGE-style routing	Gated shared/specialized experts	Safe-anchor routing for reliability, evaluated no-label

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